# EXPERIMENTAL EVALUATION

## Isolation Forest for Ethereum Wallet Anomaly Detection

Consolidated findings from experiments B1-B22

**WORKING MODEL: B6b | 20 FEATURES | 300 TREES**

Dataset: 10,000 rows and 23 numeric features

### Purpose

Document feature-selection, stability, sensitivity, consensus and clustering experiments; identify a defensible working configuration; and state the boundaries of the available evidence.

**Interpretation note: this is an unsupervised anomaly-detection study. No fraud labels or wallet identifiers are present in the analysed CSV.**

# Executive Summary

This report consolidates the Isolation Forest experiments B1-B22 on a preprocessed Ethereum wallet feature dataset containing 10,000 rows and 23 numeric features. Experiments examined alternative feature sets, repeated held-out rankings, permutation sensitivity, cross-configuration consensus, and exploratory clustering of repeatedly high-ranked anomaly candidates.

**Main decision**

Retain B6b (20 features) as the working Isolation Forest configuration. It demonstrated strong ranking consistency against the full-feature baseline, was competitive across alert budgets, and was nearly unchanged by robust scaling. This is a model-selection decision based on consistency and simplicity, not verified fraud-detection accuracy.

### Headline findings

- B6b achieved the highest mean pairwise top-5% Jaccard similarity among the four configurations compared in B6: 0.8689; its mean pairwise Spearman correlation was 0.9902.
- The B9 scaling experiment showed near-identical rankings for unscaled and robust-scaled B6b: Spearman 0.99987 and mean top-5% Jaccard 0.97566.
- A repeated held-out consensus procedure identified 94 rows meeting the strong-consensus criterion across the 23-, 20- and 18-feature configurations.
- Permutation sensitivity highlighted min value received, current_eth_balance, and the ERC-20 received-diversity group as influential to ranking behaviour.
- Clustering of the 94 candidates produced a three-cluster solution of 77, 14 and 3 rows, but its interpretation was sensitive to feature representation and one feature (min value received).
- No result establishes fraud precision, recall, F1, ROC-AUC, or confirmed wallet identities because the analysed dataset has no verified fraud labels or wallet identifiers.

# 1. Dataset and Scope

The source file was D2_IsolationForest_preprocessed_23_features.csv. It contains 10,000 rows and 23 numeric features. The input appeared preprocessed or transformed; feature values should not be interpreted as raw ETH units without checking the preprocessing pipeline.

### 1.1 Important data limitations

- No verified fraud labels are present in this CSV.
- No wallet identifiers are present, so row indices cannot independently be matched to real Ethereum addresses.
- The dataset contains 458 rows belonging to exact duplicate-feature groups, including 382 extra duplicate copies. The 94 final cross-configuration consensus candidates did not belong to such groups.
- Endpoint concentrations are notable: Sent tnx = 100 for 31.36% of rows, Received Tnx = 100 for 29.62%, and total transactions = 200 for 20.36%. Capping or data-collection limits are possible explanations, but are not confirmed.

# 2. Methodology

### 2.1 Isolation Forest configuration

The principal experiments used Isolation Forest with 300 trees, contamination="auto", parallel processing and fixed random seeds for reproducibility. Anomaly score was computed as the negative of score_samples, so that higher values mean more anomalous. The working B6b model uses 20 features and no scaling.

### 2.2 Feature configurations

| Variant | Features | Definition |
|---|---|---|
| A | 23 | Full-feature baseline. |
| B1 | 20 | Removes max value received, max val sent, and ERC20 max val rec. |
| B6a | 18 | Removes the three maximum-value features plus ERC20 uniq rec token name and Received Tnx. |
| B6b | 20 | Removes Sent tnx, Received Tnx, and total transactions (including tnx to create contract). |
| B9a | 16 | B6b with four additional selected features removed: max value received, max val sent, ERC20 max val rec, and ERC20 uniq rec token name. |
| B9b | 20 | B6b with RobustScaler applied. |
| B12 | 18 | B6b without avg val received and max value received. |

### 2.3 Evaluation measures

- Spearman rank correlation: agreement in overall anomaly ordering.
- Top-k overlap and Jaccard similarity: agreement in the selected highest-priority anomaly candidates.
- Permutation sensitivity: ranking changes when one feature or a correlated feature group is shuffled in held-out data.
- Adjusted Rand Index (ARI) and silhouette score: exploratory clustering agreement and geometric separation.
- Runtime: descriptive execution time; not a measure of fraud-detection quality.

## 3. Compact Results Summary (B1-B22)

The table below condenses the central result from each experiment group. Full interpretation follows in Section 4.

| Experiments | Focus | Key result | Interpretation |
|---|---|---|---|
| B1-B6 | Feature selection and seed stability | B6b pairwise top-5% Jaccard 0.8689; Spearman 0.9902. | B6b selected as working model. |
| B7-B9 | Held-out ranking and scaling | A vs B6b: 89.4/100 average top-5% overlap; scaled vs unscaled B6b Spearman 0.99987. | Rankings broadly agree; scaling adds little. |
| B10-B11 | Feature sensitivity | min value received had largest individual Spearman drop (0.05264); ERC-20 received-diversity group drop 0.12533. | Feature values and correlated groups influence rankings. |
| B12-B13 | Group ablation and alert budgets | Removing all four groups: 59.04/100 mean top-5% overlap vs B6b. B6b most seed-stable at 1%, 5%, 10% budgets. | Aggressive feature reduction changes the shortlist. |
| B14-B15 | Consensus candidates | 94 candidates shared across A, B6b and B12 under the exploratory criterion. | A cross-configuration shortlist was established. |
| B16-B17 | Candidate profile and data quality | 60/94 had max value received above background P95; 59/94 had total ether received above P95. No candidate row belonged to an exact duplicate group. | Many candidates have high transaction-value and ERC-20 diversity features. |

| Experiments | Focus | Key result | Interpretation |
|---|---|---|---|
| B18-B20 | Exploratory clustering | For k=3: sizes 77, 14, 3; silhouette 0.67366. B20 cross-feature-set ARI = 1.0 for k=2 and k=3. | Stable under tested feature-set changes, but the smallest cluster is tiny. |
| B21-B22 | Cluster sensitivity | Removing min value received: ARI 0.11011 (k=2) and 0.14439 (k=3). Alternative transformations substantially altered partitions. | Cluster interpretation is feature- and transformation-sensitive. |

**Overall result**

The work supports a reproducible unsupervised anomaly-ranking workflow with B6b as the working configuration. It identifies 94 cross-configuration consensus candidates and describes their feature profiles; it does not confirm fraud or demonstrate fraud-classification accuracy.

# 4. Detailed Experimental Findings

## 4.1 B1-B6: Feature Selection and Seed Stability

The initial experiments compared the original 23-feature configuration with reduced variants. B1 removed three maximum-value features and retained highly correlated overall rankings. B2 evaluated B1 over five seeds: mean pairwise top-5% Jaccard was 0.8471 and mean pairwise Spearman correlation was 0.9876. B3 compared the full baseline and B1 over five seeds; A had pairwise top-5% Jaccard 0.8381, while B1 had 0.8471. This difference was small and did not indicate fraud accuracy.

B4 identified 366 rows that remained in the top 5% across the ten runs covering A and B1. B5 investigated those candidates and found none belonged to an exact duplicate-feature group. It also revealed strongly correlated value and address-diversity features. B6 compared four feature sets: A, B1, B6a and B6b. B6b had the strongest pairwise top-5% stability (0.8689) and pairwise Spearman correlation (0.9902) among those variants, motivating its selection as the working configuration.

## 4.2 B7-B9: Held-Out Rankings and Scaling

B7 used a fixed 80/20 development/held-out split. A and B6b shared 88 of their 100 highest-ranked held-out rows (Jaccard 0.78571). B8 repeated evaluation over five split seeds and five model seeds. Matched comparisons showed an average of 89.4 shared top-5% rows between A and B6b (mean Jaccard 0.80923). These experiments suggest substantial ranking agreement across configurations, but do not establish which model better identifies fraud.

B9 found that RobustScaler scarcely changed B6b rankings: mean Spearman was 0.99987 and mean top-5% Jaccard 0.97566 between scaled and unscaled B6b. B9a, which removed additional correlated features, had lower agreement with the baseline. There was therefore little reason to use scaling solely to alter Isolation Forest rankings.

## 4.3 B10-B11: Feature Sensitivity

B10 permuted individual features on held-out rows without refitting the trained model. min value received had the largest mean Spearman drop (0.05264). current_eth_balance caused the strongest change in top-5% membership, with mean Jaccard 0.74659. B11 shuffled correlated feature groups together. The group containing ERC20 uniq rec addr and ERC20 uniq rec token name had mean Spearman drop 0.12533 and mean top-5% Jaccard 0.65946. These values indicate sensitivity of model rankings to the input columns, not causal influence or fraud relevance.

## 4.4 B12-B13: Group Ablation and Alert Budgets

B12 removed correlated feature groups. Removing all four groups reduced the model to 12 features and lowered mean top-5% overlap with B6b to 59.04 rows; mean Spearman was 0.91041. Runtime was nearly unchanged (approximately 0.916 seconds for the 12-feature version versus 0.923 seconds for B6b). B13 investigated 1%, 2%, 5% and 10% alert budgets across seeds. B6b was most stable at 1%, 5% and 10%; the 18-feature B12 variant was most stable at 2%. This supports B6b as a general working model, not as a universal winner for every alert budget.

## 4.5 B14-B17: Consensus and Behavioural Profiles

B14 aggregated repeated held-out rankings for B6b. It identified 117 candidates appearing in the held-out top 5% in at least 80% of their evaluations, with minimum eligibility requirements. B15 then assessed agreement across A, B6b and B12, yielding 94 candidates that met the criterion for all three configurations. The threshold was exploratory, and repeated evaluations overlap.

B16 compared the 94 candidates with the remaining 9,906 rows. Candidate medians were higher for current_eth_balance (0.11498 versus 0.00005), ERC20 uniq rec token name (46.5 versus 3), avg val sent (7.91129 versus 0.69294), avg val received (8.39666 versus 0.91959), and ERC20 uniq rec addr (45.5 versus 4). No candidate belonged to an exact duplicate-feature group. Because the candidates were selected using the same features, these contrasts are descriptive and selection-dependent.

B17 assessed each candidate feature relative to the background distribution. Values above the background 95th percentile appeared most often in max value received (60 candidates), total ether received (59), max val sent (55), and total Ether sent (55). These results indicate that transaction-value and transfer features dominate the descriptive profile of the consensus candidates. They do not establish a fraud typology.

## 4.6 B18-B20: Exploratory Clustering

K-means clustering was performed on the 94 consensus candidates after RobustScaler transformation of B6b features. The three-cluster solution achieved silhouette 0.67366 and cluster sizes 77, 14 and 3; the two-cluster solution achieved silhouette 0.64611 with sizes 78 and 16. Four- and five-cluster solutions had lower silhouettes. The three-cluster solution was retained for exploratory analysis, but the three-member cluster is too small to justify a separate fraud category.

B19 reported mean ARI 0.99750 for k=3 under 80% candidate subsampling, with mean cluster-best Jaccard 0.99618. However, the smallest cluster did not reproduce perfectly in every repeat (minimum matched Jaccard 0.50). B20 found identical partitions across the 20-, 18- and 16-feature configurations for both k=2 and k=3 (cross-feature-set ARI 1.0). These results suggest stability under the specific tested feature changes, but later results show that this structure is not invariant to all transformations.

## 4.7 B21-B22: Feature and Transformation Sensitivity

B21 removed one feature at a time from the candidate set used for clustering. Removing min value received changed the clustering substantially: mean ARI was 0.11011 for k=2 and 0.14439 for k=3. Removing ERC20 min val rec had a smaller effect for k=3 (ARI 0.87963), while removing most other individual features preserved the reference partition. Although removing min value received increased the k=2 silhouette score to 0.79809, it also produced strongly imbalanced clusters averaging approximately 4 and 90 candidates. A higher silhouette score alone does not guarantee a more useful partition.

B22 evaluated RobustScaler, StandardScaler, Yeo-Johnson, quantile-to-normal transformation, and winsorised RobustScaler. For the full B6b feature set, RobustScaler retained the reference partitions, while winsorised RobustScaler retained them closely (ARI 1.0 for k=2 and 0.95831 for k=3). StandardScaler, QuantileNormal and Yeo-Johnson substantially changed the partitions. This indicates that the cluster interpretation depends on representation choices. The results concern clustering of the selected candidates, not the trained Isolation Forest detector.

# 5. Current Working Model

| Item | Working choice |
|---|---|
| Algorithm | Isolation Forest |
| Feature set | B6b: 20 features |
| Trees | 300 |
| Contamination | auto |
| Random seed | 42 for the frozen reproducible model |
| Scaling | None; B9 showed negligible ranking effect from RobustScaler |
| Score convention | Negative score_samples; higher = more anomalous |
| Intended role | Unsupervised behavioural anomaly ranking / candidate prioritisation |

## B6b features retained

- Time Diff between first and last (Mins)
- Total ERC20 tnxs
- Unique Received From Addresses
- ERC20 uniq rec addr
- ERC20 uniq rec token name
- total ether received
- avg val received
- Avg min between received tnx
- max value received
- ERC20 min val rec
- min value received
- total Ether sent
- ERC20 avg val rec
- ERC20 max val rec
- Avg min between sent tnx
- max val sent
- avg val sent
- Unique Sent To Addresses
- ERC20 uniq sent token name
- current_eth_balance

# 6. Limitations and Validity

- No verified fraud labels: precision, recall, F1, PR-AUC and ROC-AUC cannot be calculated for this dataset.
- No wallet identifiers: row indices cannot be independently mapped to Ethereum addresses.
- Repeated splits overlap: multi-seed results measure sensitivity, not independent replication across separate datasets.
- Candidate-selection bias: the same input features influence both anomaly selection and the subsequent descriptive feature profiles and clustering.
- Representation sensitivity: B22 demonstrated that alternative transformations substantially change the candidate clustering.
- No confirmed fraud typologies: behavioural clusters describe statistical profiles only and must not be labelled as fraud categories without external evidence.

- The current work does not demonstrate adversarial robustness, temporal generalisation, or performance at million-wallet scale.

# 7. Conclusion and Next Steps

The B1-B22 sequence establishes a reproducible exploratory anomaly-ranking workflow and supports B6b as the working Isolation Forest configuration. B6b offered strong ranking agreement with the full-feature baseline, was little affected by RobustScaler, and provided a stable basis for selecting a cross-configuration consensus shortlist. The 94 candidates showed elevated transaction-value and ERC-20 receiving-diversity features relative to the background population.

The experiments do not establish that the candidates are fraudulent, and they do not identify real wallets. The clustering analyses are exploratory and representation-sensitive. The next scientific priority is to obtain verified fraud labels and wallet identifiers, validate the preprocessing pipeline, and evaluate the detector against independent evidence. A chronologically separated evaluation would be preferable if timestamps and raw transaction histories are available.

**Appropriate claim**

The present system is an unsupervised anomaly detector and candidate-prioritisation component. It is not yet a validated Ethereum fraud classifier.

# Appendix A. Interpretation Notes

- Top-5% means the highest-scoring 5% of a specified evaluated set; it is an investigation budget, not a fraud rate.
- Jaccard similarity measures overlap of selected row sets; Spearman measures similarity of rankings.
- Anomaly percentile is relative to a reference distribution; it is not a calibrated fraud probability.
- A high silhouette score describes geometric separation in a chosen feature space, not external validity of a behavioural category.
- Permutation effects measure ranking sensitivity and can be affected by feature correlation and unrealistic shuffled combinations.
- Values are shown in the dataset representation as supplied. Raw-unit interpretation requires the documented preprocessing recipe.